Chapter 3 Conditional Probability & Independence

① If $(P(F)) > 0$ then,

$$\boxed{P(E|F) = \frac{P(E \cap F)}{P(F)}}$$

② The multiplication rule

$$P(E_1, E_2\, E_3 \ldots\ldots E_n) = P(E_1) \cdot P(E_2|E_1) \cdot P(E_3|E_1 \cap E_2) \ldots\ldots P(E_n | E_1 \ldots\ldots E_{n-1})$$

$P(E_3|E_1 \cdot E_2)$ E_3 occurs given both E_1 & E_2 occurs.

③ Baye's formula.

$$\boxed{P(F|E) = \frac{P(E|F) \cdot P(F)}{P(E)}}$$

$$\begin{aligned} P(E) &= P(EF) + P(EF^c) \\ &= P(E|F) \cdot P(F) + P(E|F^c) P(F^c) \\ &= P(E|F) \cdot P(F) + P(E|F^c)[1 - P(F)] \end{aligned}$$

④ odds of an event A are defined by..

$$\boxed{\frac{P(A)}{P(A^c)} = \frac{P(A)}{1-P(A)}}$$

Consider an Hypothesis that H is true with Prob. $P(H)$ & evidence E is introduced. The the Condnal Probability that H is true & isn't given by.

$$\boxed{P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}}$$

$$\boxed{P(H^c|E) = \frac{P(E|H^c)\,P(H^c)}{P(E)}}$$

$$\boxed{\frac{P(H|E)}{P(H^c|E)} = \frac{P(H)}{P(H^c)}\,\frac{P(E|H)}{P(E|H^c)}}$$

Events EF_i, $i=1,\ldots n$ are mutually exclusive then.

$$\boxed{\begin{aligned} P(E) &= \sum_{i=1}^{n} P(EF_i) \\ &= \sum_{i=1}^{n} P(E|F_i)\,P(F_i) \end{aligned}}$$

formula is used below.

Proposition 3.1

$$\boxed{P(F_j|E) = \frac{P(E|F_j)\,P(F_j)}{\sum_{i=1}^{n} P(E|F_i)\,P(F_i)}}$$

⑤ Independent events :-

2 Events E & F are said to be independent if $\boxed{P(EF) = P(E) \cdot P(F)}$ holds true.

3 events are said to be independent if :-

$$\boxed{\begin{aligned} P(EFG) &= P(E) \cdot P(F) \cdot P(G) \\ P(EF) &= P(E) \cdot P(F) \\ P(F \cdot G) &= P(F) \cdot P(G) \\ P(E \cdot G) &= P(E) \cdot P(G) \end{aligned}}$$

for more events

$$P(E_1, E_2, E_3, E_4 \ldots\ldots E_r) = P(E_1') \cdot P(E_2') \cdot P(E_3') \ldots\ldots P(E_r')$$

6) The binomial Probability formula.

$$\boxed{P(\text{Exactly } k \text{ Success}) = \binom{n}{k} p^k (1-p)^{(n-k)}}$$

(a) eg Parallel system

$P(\text{system works}) = 1 - P(\text{all fail})$

Independence $P(\text{all fail}) = (1-P_1)(1-P_2)(1-P_3)(\ldots\ldots)(1-P_n)$

$$\boxed{P(\text{works}) = 1 - \prod_{i=1}^{n} (1-P_i)}$$

7) Gambler's Ruin.

$$P_i = \begin{cases} \dfrac{(1-(q/p)^i)}{1-(q/p)} P_r & \text{if } q/p \neq 1 \\ 1-(q/p) & \text{if } q/p = 1 \end{cases}$$

fact $P_N = 1$

$$P_i = \begin{cases} \dfrac{1-(q/p)^i}{1-(q/p)^N} & \text{if } p \neq 1/2 \\ 1/N & \text{if } p = 1/2. \end{cases}$$

Hence.

$$P_i = \begin{cases} \dfrac{1-(q/p)^i}{1-(q/p)^N} & \text{if } p \neq 1/2 \\ i/N & \text{if } p = 1/2. \end{cases}$$

E = Event A ends ~~ends~~ up with all the money

i = initial Amt he starts with.

$P_i = P(E)$

Q_i = Prob. B winds up with all the money A starts with i

$$Q_i = \begin{cases} \dfrac{1-(p/q)^{N-i}}{1-(p/q)^N} & \text{if } q \neq 1/2 \\[2ex] \dfrac{N-i}{N} & \text{if } q = 1/2. \end{cases}$$

8 $P(\cdot|F)$ is a Probability

Prop 5.1 (a) $0 \le P(E|F) \le 1$

(b) $P(S|F) = 1$

(c) if E_i, $i = 1,2,3,\dots$ are mutually exclusive events then.

$$\boxed{P\left(\bigcup_{i=1}^{\infty} E_i|F\right) = \sum_{i=1}^{\infty} P(E_i|F)}$$

equivalent to

$$P(E_1|F) = P(E_1|E_2F) \cdot P(E_2|F) + P(E_1|E_2^cF)\, P(E_2^c|F)$$